Finite-Gradient Conversion and Conserving Neutrino-Metric Matching in WRRA

MCC 3.0 Item 6 follow-up, working v0.4

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Verified inputs → WRRA transformation → outputs → falsification

Verified inputs. Retain v0.3's fixed reservoir fraction $\beta = 7.62452468358126e-07$, proper pulse, starting scale, background normalization and 128-cell SOURCE seed. Split the spectator radiation using the inherited $N_{\text{eff}} = 3.044$ prescription, giving $f_\nu = 0.408743970116$. The free-streaming/perfect-fluid content and reservoir velocity transfer are new disclosed effective-model choices.

WRRA transformation. Evolve the complete finite-k effective transition state rather than only its local zero-gradient entropy. Pair source and recipient energy and momentum, evolve metric and neutrino hierarchy, match by conserving the total stress-energy, then continue the matched state.

Outputs. 643 new finite checks pass; all 64 original modes are evolved and matched. The zero-gradient coefficients reproduce v0.3. New complete species states and their action on the actual seed are exported.

Falsification. Failed conservation, Einstein constraints, matching continuity, convergence or zero-gradient recovery require revision. The current result is not a newly fitted CMB angular spectrum.

Equations and variables

Use $8\pi G_{\text{N}}/3 = 1$, the relative scale factor $a(0) = 1$ and $H(0) = 1$. Dots below denote dimensionless local proper time; primes denote conformal time. The physical starting scale remains $a_i = 10^{-10}$. Radiation budgets are $C = \rho_x a^4, G = \rho_\gamma a^4, V = \rho_\nu a^4$ and $M = \rho_m a^3$. Initially $C = \beta, G = (1-\beta)(1-f_\nu)$, $V = (1-\beta)f_\nu, M = 0$.

The disclosed transfer prescription is

$$Q^\mu_m = -Q^\mu_x = \Gamma(T)(1+\xi)\rho_x u^\mu_x.$$

Here T is the declared geodesic-reference proper-time clock, with conversion active for $0 < T < 1$. The synchronous slice sets its perturbation to zero. Under another gauge the clock perturbation must also transform; it is not a universal absolute-time assertion. The scalar modulation ξ is the frozen WRRA seed, conserved to first order. Its unit-amplitude response is evolved and subsequently multiplied by the actual small seed. This constitutive choice is an explicit model input, not a particle reaction derived from a microscopic Hamiltonian.

The inherited background is

$$\dot N = H, \quad \dot C = -\Gamma C, \quad \dot M = \Gamma C/a,$$
$$\dot H = -\tfrac{1}{2}[4(C+G+V)/a^4 + 3M/a^3], \qquad \dot W = H(C+G+V)/a.$$

Let $d_i = a^{3(1+w_i)}\delta\rho_i, P_i = a^4(\rho_i + p_i)\theta_i/K^2, L = \dot h$, and $P = \sum_i P_i$. There is no division by the initially zero matter density during integration. In the unit rate-modulation response,

$$\dot d_x = -K^2 P_x/a - 2CL/3 - \Gamma(d_x + C),$$
$$\dot d_\gamma = -K^2 P_\gamma/a - 2GL/3, \quad \dot d_\nu = -K^2 P_\nu/a - 2VL/3,$$
$$\dot d_m = -K^2 P_m/a^2 - ML/2 + \Gamma(d_x + C)/a.$$

The momentum equations retain the receiver source:

$$\dot{P}_x = d_x/(3a) - 3\Gamma P_x/4, \quad \dot{P}_m = 3\Gamma P_x/4,$$
$$\dot{P}_\gamma = d_\gamma/(3a), \qquad \dot{P}_\nu = d_\nu/(3a) - 4VK^2 s_\nu/(3a).$$

For a numerically stable $K \to 0$ limit, use $s_\nu = \sigma_\nu/K^2$, $f_\ell = F_{\nu\ell}/K^\ell$ and $g = (h' + 6\eta')/K^2$:

$$\dot{\eta} = 3P/(2a^3), \quad L = K^2 g/a - 6\dot{\eta},$$
$$\dot{g} = -2Hg + 2\eta/a - 12Vs_\nu/a^3,$$
$$\dot{s}_\nu = [4v_\nu/15 + 2g/15 - 3K^2 f_3/10]/a,$$
$$\dot{f}_\ell = [\ell f_{\ell-1} - (\ell+1)K^2 f_{\ell+1}]/[(2\ell+1)a], \quad \ell \geq 3,$$

where $v_\nu = P_\nu/(4V/3)$ and $f_2 = 2s_\nu$. The upper hierarchy is set to zero beyond the stated truncation. Convergence is tested at $\ell_{\max} = 16, 24$ together with halved time steps; no claim is made for arbitrary wavenumber or arbitrarily long propagation.

The Hamiltonian constraint is inspected without resetting a metric variable:

$$K^2\eta - \tfrac{1}{2}a^2 HL + \tfrac{3}{2}a^2\delta\rho = 0.$$

An independent implementation evolves L through the Einstein trace equation,

$$\dot{L} = -3HL + 2K^2\eta/a^2 - 9\delta p,$$

and unscaled neutrino multipoles. Its trajectory is compared against the shear-based implementation. The energy/momentum residuals also directly inspect cancellation of the transfer terms; they are consistency checks, not independent observational tests.

Executed matching operation

At $T = 10$, retain $(h, \eta, g, M, d_m, P_m)$ and every neutrino moment. Absorb the remaining perfect reservoir radiation into the perfect photon-like bath:

$$G^+ = G^- + C^-, \quad d_\gamma^+ = d_\gamma^- + d_x^-, \quad P_\gamma^+ = P_\gamma^- + P_x^-; \qquad C^+ = d_x^+ = P_x^+ = 0.$$

This preserves background/perturbed total energy, pressure, momentum and anisotropic stress on the same proper-time slice, hence the induced metric and its first derivative. The operation is an exact linear stress-energy regrouping of two perfect radiative components. It is not a calculated microscopic thermalization or baryon-production reaction. Both the unmerged and merged systems are actually continued to $T = 30$. Their agreement tests this regrouping only, not equality of every possible physical post-transition model.

The export retains separate photon and neutrino densities, velocities and the full computed neutrino hierarchy, rather than projecting it down to the two v0.3 primordial coefficients. Photon-reservoir absorption is a disclosed branch choice. Other destinations require a separately conserving matching rule.

Numerical outputs

Quantity	Result
New finite calculation checks	643 passed
Original finite wavenumbers actually evolved and matched	64
Zero-gradient entropy response	0.723567385056114
Zero-gradient radiation curvature response	-6.60614975285515e-08
Largest physical $k/(aH)$ at matching	0.000679763743809
Mode-64 entropy-response relative change from v0.3	9.10792996578e-10
Mode-64 radiation-curvature relative change	-6.31401660955e-08
Actual 128-cell entropy field relative L2 change	8.81487192574e-11
Largest matched/unmerged continuation difference, declared norm	9.75856e-16
Largest independent-metric state difference, declared norm	3.89482e-11

At the largest original mode, per unit rate modulation, $S_{mr} = 0.723567385715136$ and $\zeta_r = -6.60614933574178e-08$. The actual seed gives entropy RMS=1.78372913373e-08. No amplitude was refitted. Printed digits specify reproducible outputs rather than empirical precision.

The merge also produces a branch-specific photon/neutrino entropy $S_{\nu\gamma} = 3.35192239644e-07$ per unit modulation at mode 64. It is multiplied by the small actual seed; it is not a measured cosmological isocurvature amplitude. This separate species memory is retained in the export.

Residual across the original 64 modes	Maximum declared normalized value
Einstein Hamiltonian	2.09149e-14
Einstein trace	9.27668e-15
Perturbed energy conservation	3.35395e-14
Perturbed momentum conservation	5.68e-15

Each normalizer is explicit in `sample_audit` and in the negative-control JSON. These are numerical residuals in the declared units, not measurement uncertainties. The momentum constraint is used to evolve η; its identical re-evaluation is not counted as an independent test. The Hamiltonian and independent trace trajectories provide additional checks.

Stress cases extend to $k/(aH) = 4.58257$ at matching. They are diagnostic wavenumbers outside the original finite SOURCE grid, not new universe-wide source modes. The comparison removes neutrino shear to define a separate perfect-fluid model; its metric differs. The combined hierarchy/time-step convergence changes the selected metric/curvature responses by at most 2.54749e-12 in the declared norm.

Rejecting controls

1. Delete the recipient momentum source while leaving energy exchange intact: energy residual remains small, but normalized Hamiltonian residual becomes 0.0155773, trace residual 1.63109, and momentum residual 4.3044. Energy-only auditing would miss the error.
2. Drop the reservoir density perturbation at matching: the relative total-density jump is 0.0733204.
3. Drop its momentum: the relative total-momentum jump is 0.62217.

The 643 checks comprise 576 original-mode state/constraint/matching checks; three zero-gradient checks; five finite-gradient/hierarchy response checks; 21 actual-seed field checks; 35 stress-case matching/independent-solver/convergence checks; and three rejecting controls. Previous v0.3 checks are not added to this count.

Falsification conditions and scope

Failure to recover the fixed v0.3 zero-gradient outputs, a nonconvergent hierarchy/time integration, failed Einstein constraints, unpaired transfer momentum, or a density/momentum/geometry jump on the declared matching surface rejects or revises this branch. The frozen SOURCE is used without a new observational amplitude fit. A later observational comparison must use the completed state and its declared physical history.

Implemented: the effective conversion interval now has finite spatial gradients at linear amplitude, separately moving components, a dynamical synchronous metric, a collisionless massless-neutrino hierarchy, and an explicitly executed conserving matching/continuation. The small-gradient limit is computed rather than imposed as the final state.

Still open: injecting this entire state into the production CAMB initial callback and computing new T/E/lensing spectra; a self-consistent nonzero baryon background/production and tight-coupled photon-baryon system in the conversion interval; massive-neutrino momentum dependence, collisional neutrino decoupling, a MeV thermal history, microscopic transfer rates, finite-quantum event timing, nonlinear dynamics and new observational likelihoods. A massless collisionless neutrino bath and a perfect photon-like bath are explicit effective inputs during this experiment, not a derived early-Universe thermal history. This extends Item 6 but does not close all of its physical branches.

The inherited v0.3 full CMB spectra are not relabeled as v0.4 predictions. No new CMB angular spectrum or new observational detection is claimed here. Standard GR/Boltzmann equations are disclosed construction material; their present integration is limited to the equations actually evolved.

Reproduction and primary sources

Python 3.12, NumPy 2.3.5 and SciPy 1.17.0. Run `python run_checks.py`; it needs only the two frozen files in `inputs/`. `finite_gradient.py` and `independent_metric.py` contain both metric formulations. `results/audit.json` contains every named test, absolute residuals for counterexamples and normalization information. `matching_states.json` provides before/after species states and velocity units; `all_64_transition_responses.npz` stores normalized responses; `actual_128_cell_matched_fields.npz` applies them to the actual seed. `make_reports.py` additionally uses Matplotlib and Pandoc/XeLaTeX to rebuild the reports. A clean numerical rerun requires no CAMB binary or network.

1. Choi, W. and Choi, J. v0.3 predecessor, DOI 10.5281/zenodo.23489547.
2. Ma, C.-P. and Bertschinger, E. Cosmological Perturbation Theory in the Synchronous and Conformal Newtonian Gauges, equations (21), (49), astro-ph/9506072.
3. Malik, K. A. and Wands, D. Cosmological perturbations, interacting-fluid conservation and transfer, arXiv:0809.4944.

External sources specify the adopted field equations; the reservoir prescription, finite-source response, matching operation and numerical outputs are the declared WRRA construction executed here. Authors: Wonsik Choi and Jeongin Choi, independent researchers. AI assistance: implementation, computational checks and language preparation. Original new material: CC BY 4.0. No new Zenodo/GitHub publication is asserted by this working report.

Diagnostic plots

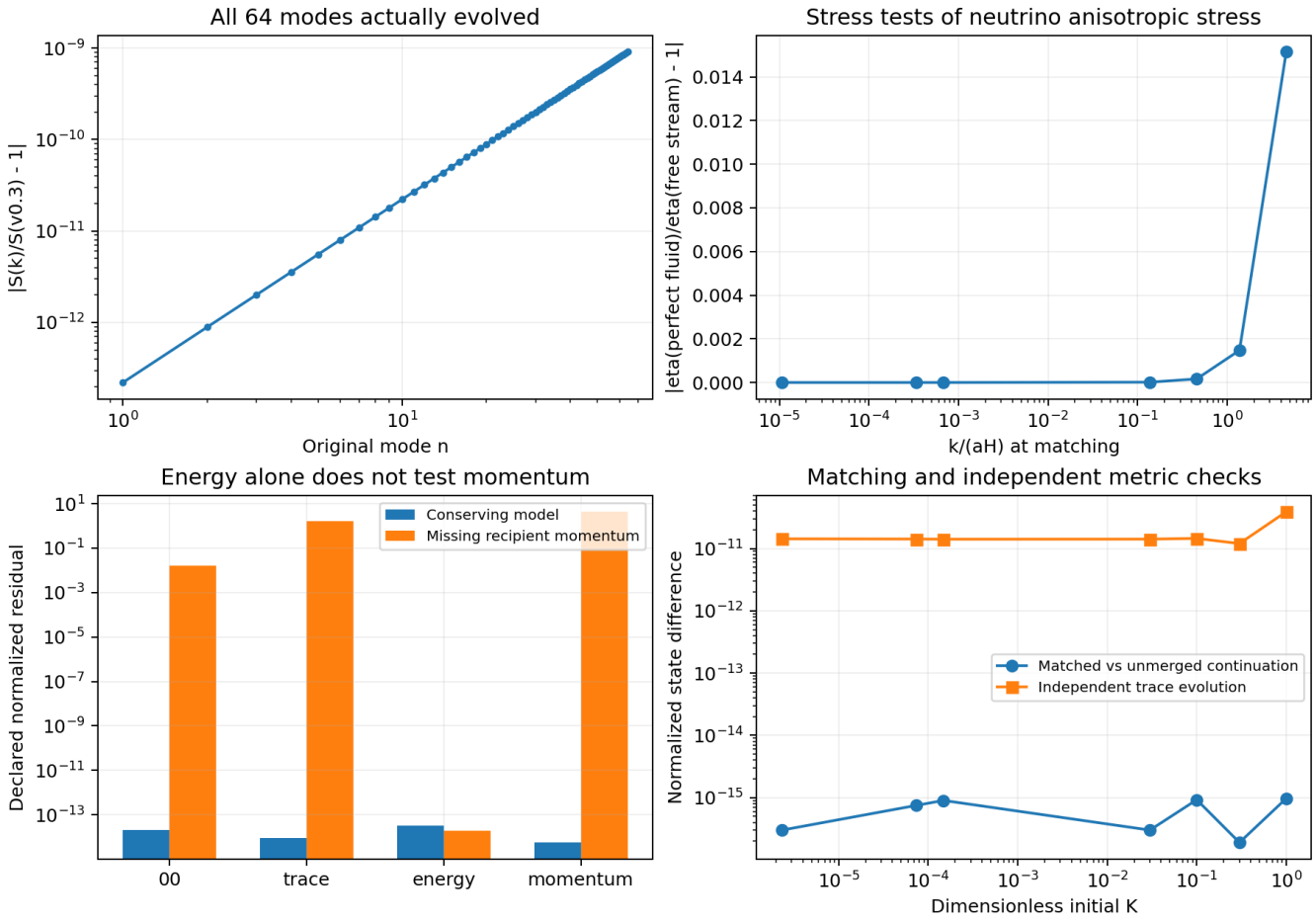


Figure 1: Finite-gradient corrections, neutrino-stress comparison, conservation counterexample and matching tests.